

RESEARCH ARTICLE

Development of machine learning algorithms to predict achievement of minimal clinically important difference for the KOOS-PS following total knee arthroplasty

Akhil Katakam^{1,2}  | Aditya V. Karhade¹ | Austin Collins¹ | David Shin¹ | Charles Bragdon¹ | Antonia F. Chen³  | Christopher M. Melnic^{1,2} | Joseph H. Schwab¹ | Hany S. Bedair^{1,2}

¹Department of Orthopaedic Surgery, Massachusetts General Hospital, Harvard Medical School, Boston, Massachusetts, USA

²Department of Orthopaedic Surgery, Newton-Wellesley Hospital, Newton, Massachusetts, USA

³Department of Orthopaedic Surgery, Brigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts, USA

Correspondence

Hany S. Bedair, Department of Orthopaedic Surgery, Massachusetts General Hospital, Harvard Medical School, 55 Fruit St, Boston, MA 02114, USA.
Email: HBEDAIR@mgh.harvard.edu

Abstract

As cost-effective measures become increasingly implemented in the US healthcare system, changes in patient-reported outcome measure (PROM) scores can be utilized to indicate patient satisfaction following procedures including total knee arthroplasty (TKA). The primary aim of this study was to develop and evaluate machine learning algorithms to predict achievement of the minimal clinically important difference (MCID) for the Knee Injury and Osteoarthritis Outcome Score-Physical Function Short Form (KOOS-PS) at 1-year following TKA. A retrospective review of primary TKA patients between 2016 and 2018 was performed. Variables considered for prediction included demographics and preoperative PROMs. The KOOS-PS MCID was calculated via a distribution-based method. Five machine learning algorithms were developed and tested by discrimination, calibration, Brier score, and decision curve analysis. Among the 744 patients who met the inclusion criteria, 385 (72.8%) patients achieved the MCID. The elastic-net penalized logistic regression model was selected as the best performing model (c-statistic 0.77, calibration intercept -0.02, calibration slope 1.15, and Brier score 0.14). The most important variables for MCID achievement were preoperative KOOS-PS score, preoperative VAS Pain, preoperative opioid use, preoperative PROMIS global mental health score, age, and sex. Algorithms were incorporated into an open-access digital application available at https://sorg-apps.shinyapps.io/tka_koos_mcid/. This study is the first to predict the probability of achieving the KOOS-PS MCID following TKA using a machine learning-based approach. The results were used to develop a clinical decision aid based on commonly collected predictive variables to preoperatively predict an individual patient's likelihood of attaining an acceptable outcome following TKA.

KEYWORDS

KOOS, machine learning, MCID, TKA

1 | INTRODUCTION

Patient-reported outcome measures (PROMs) provide a subjective assessment of well-being that cannot be captured by traditional objective clinical outcomes such as infection rates, readmissions, and complications. As value-based health care and cost-effectiveness measures gain traction in the US healthcare system, changes in preoperative and postoperative PROM scores can be utilized to indicate the efficacy of a procedure as identified by the patient. Identifying patients that may not achieve the minimal clinically important difference (MCID), or the minimal amount of change in a PROM to obtain a meaningful outcome, may assist in preoperative planning, patient counseling, timing of intervention, and lead to more stringent postoperative monitoring. Across orthopedics, there have been numerous studies that have used traditional statistical models to predict achievement of the MCID in patients with total joint arthroplasty (TJA); however, there have been fewer such studies that have applied machine learning to answer the question.^{1,2}

Machine learning is a data analysis method that combines computer science and statistical techniques that is being increasingly used as a prediction and modeling tool within the field of orthopedics.^{3–5} A similar methodology has been used to develop preoperative tools that can predict the likelihood of postoperative opioid use in both orthopedic spine surgery and TJA.^{4,6,7} Although there exist previous studies analyzing the capabilities of machine learning in predicting MCID achievement in patients with TJA, this study is the first to apply the methods specifically to total knee arthroplasty (TKA) and the Knee Injury and Osteoarthritis Outcome Score-Physical Function Short Form (KOOS-PS).^{1,2}

The primary aim of this study was to develop and evaluate machine learning algorithms to predict achievement of the MCID for the KOOS-PS at 1-year after TKA and provide explanations of these predictions to aid interpretability of machine learning models.

2 | METHODS

2.1 | Guidelines

This retrospective, Level III study was conducted with Institutional Review Board approval and in accordance with the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis guidelines and the Guidelines for Developing and Reporting Machine Learning Predictive Models in Biomedical Research.^{8,9}

2.2 | Patients and variables

Among patients undergoing primary unilateral TKA in two academic medical centers and three community medical centers between January 2016 and August 2018, patients were included for analysis who were 18 years and older and who had completed at least one preoperative KOOS in the 3-months before surgery. The KOOS is a

common PROM administered to patients with TKA and assesses knee symptoms and function.¹⁰ The KOOS-PS is a 7-item physical short form derived from the daily living and sports and recreational activity subscales of the KOOS, where 0 indicates extreme difficulty and 100 indicates no difficulty. The electronic medical record was reviewed to identify the following variables: age, sex, body mass index (BMI), American Society of Anesthesiologist score, Medicaid or Medicare status, neighborhood characteristics via the United States Censuses Bureau American Community Survey zip code data (median household income, rate of high school education, and unemployment rate), preoperative laboratory values (hemoglobin, white blood cell count, platelet count, and creatinine level), preoperative opioid use, comorbidities (any Charlson comorbidity, diabetes, depression, chronic obstructive pulmonary disease, and arrhythmias), preoperative patient-reported outcomes (KOOS, Patient-Reported Outcomes Measurement Information System [PROMIS] Global physical function [PF], PROMIS Global mental health [MH], and visual analog scale for pain [VAS Pain]).

The primary outcome was an improvement of KOOS from preoperative to one-year postoperative score greater than or equal to the MCID for KOOS—success or failure to do so is hereafter referred to as “achieving the 1-year KOOS MCID.” The MCID is a tool that allows for clinically meaningful analysis of PROMs and is defined as the minimum change in PROM scores that patients perceive as beneficial or clinically meaningful.¹¹ The MCID was derived via a distribution-based method and determined to be one-half of the standard deviation of preoperative to postoperative change in KOOS among patients with postoperative follow-up beyond greater than 1-year. This methodology of obtaining the MCID has been used throughout the field of orthopedics even for KOOS and has been shown to possess excellent reliability and validity.^{12–14} Patients with less than 1-year follow-up who had reached the MCID threshold were classified as having reached MCID. Patients with less than 1-year follow-up who had not reached the MCID threshold were classified as patients with missing data for MCID.

2.3 | Missing data

Multiple imputations using the missForest methods were applied for variables missing data at less than 30%.¹⁵ These variables (rate of missing data) were: hemoglobin (21.3%), white blood cell count (21.3%), platelet count (21.3%), creatinine level (22.4%), median household income (1.5%), high school education (0.9%), unemployment rate (0.9%), preoperative PROMIS Global PF (6.3%), preoperative PROMIS Global MH (6.3%), preoperative VAS Pain (6.3%), and completing the 1-year postoperative KOOS (28.9%).

2.4 | Variable selection

The 10-fold cross-validation of random forest algorithms was used to identify the optimal subset of variables for use across the various

predictive models assessed. The 10-fold cross-validation involves splitting the data into 10 subgroups—for each of 10 models, nine subgroups of the data are used for training, and the tenth is used for testing. Random forest algorithms make repeated random selections of features, and their use in 10-fold cross-validation yields a subset of features that maintains low variance in model performance and avoids overfitting.

2.5 | Model development

Five algorithms were evaluated in this study—stochastic gradient boosting, random forest, support vector machine, neural network, and penalized logistic regression. This selection of models was based on methods outlined extensively in past studies.^{4,16} These algorithms are commonly used and are well suited for binary classification tasks, such as prediction of success/failure to achieve the MCID.²

Data were randomly split in a ratio of 70:30 between the training and validation sets of data. Models were trained using the training set data and assessed both within the training and validation set.

2.6 | Assessment of model performance

For each of the five algorithms tested in this study, model performance was evaluated using both the training set data (by 10-fold cross-validation repeated three times) and testing set data. To measure the performance of the models, the following metrics were assessed: discrimination—by the area under the curve, calibration—by both calibration intercept and calibration slope, and overall performance—by Brier score. For reference, the null model Brier score—for which the probability of meeting the 1-year KOOS MCID is assigned to be its prevalence—was calculated both in the cross-validation of the training set and in the testing set.

The best performing algorithm by these metrics was further analyzed by decision curve analysis to understand the effect of true and false positives of the model's predictions across the whole range of predicted probabilities of achieving the 1-year KOOS MCID. In addition, net benefit (the sum of weighted true and false positives) was compared between the selected algorithm and default strategies of changing management for no patients or for all patients, as well as a model based solely on preoperative KOOS.

2.7 | Application development

The best performing algorithm across the measures of model performance was incorporated into an open-access web application. The Anaconda Distribution (Anaconda, Inc.), R (The R Foundation), and Python (Python Software Foundation) were used across data analysis, model development, and web application development.

TABLE 1 Baseline characteristics of the study population, $n = 744$

Variable	n (%) median (IQR)
Age	68.0 (62.0–74.0)
Female sex	452 (60.8)
Body mass index (kg/m^2)	30.1 (26.6–34.4)
American Society of Anesthesiologist Class	
I–II	492 (66.3)
III–IV	250 (33.7)
Medicaid	55 (7.4)
Medicare	414 (55.6)
Neighborhood characteristics	
Median household income (\$)	84159 (65112–103064)
High school education (%)	23 (14–29)
Unemployment rate (%)	5.5 (4.5–7)
Laboratory values	
Hemoglobin (g/dL)	13.7 (12.8–14.5)
White blood cell ($10^3/\mu\text{L}$)	6.71 (5.66–8.06)
Platelet ($10^2/\mu\text{L}$)	239 (205–282)
Creatinine (mg/dL)	0.85 (0.72–0.99)
Preoperative opioid use	170 (22.8)
Comorbidities	
Any Charlson comorbidity	496 (66.7)
Diabetes	75 (10.1)
Depression	103 (13.8)
COPD	83 (11.2)
Arrhythmias	71 (9.5)
Preoperative patient-reported outcomes	
KOOS	56.0 (48.8–63.0)
PROMIS Global PF	39.8 (34.9–44.9)
PROMIS Global MH	50.8 (43.5–56.0)
NRS Pain	6 (5–8)

Abbreviations: COPD, chronic obstructive pulmonary disease; IQR, interquartile range; KOOS, Knee Osteoarthritis Outcome Score; MH, mental health; NRS pain, numerical rating scale for pain; PF, physical function; PROMIS, Patient-Reported Outcomes Measurement Information System.

3 | RESULTS

3.1 | Baseline characteristics and predictive factors

Approximately 744 patients undergoing primary TKA were found to meet the inclusion criteria. Median age (interquartile range [IQR]) in the cohort was 68 (62–74) years, 452 (60.8%) patients were female, and median BMI (IQR) was 30.1 kg/m^2 (26.6–34.4). For the KOOS, the median (IQR) preoperative score was 56.0 (48.8–63.0) points (Table 1). For the PROMIS Global PF, PROMIS Global MH, and the

TABLE 2 Algorithm performance on cross-validation of training set, $n = 522$, mean (95% confidence interval)

Metric	Stochastic gradient boosting	Random forest	Support vector machine	Neural network	Elastic-net penalized logistic regression
AUC	0.74 (0.71–0.77)	0.74 (0.72–0.77)	0.75 (0.72–0.77)	0.77 (0.74–0.79)	0.76 (0.74 to 0.79)
Calibration intercept	−0.01 (−0.14 to 0.12)	0.22 (0.10–0.33)	−0.28 (−0.50 to −0.05)	−0.03 (−0.16 to 0.10)	−0.02 (−0.19 to 0.14)
Calibration slope	1.04 (0.91–1.17)	0.81 (0.69–0.93)	1.26 (1.08–1.44)	1.03 (0.92–1.15)	1.07 (0.89 to 1.07)
Brier score	0.15 (0.14–0.16)	0.16 (0.15–0.16)	0.15 (0.15–0.16)	0.14 (0.14–0.15)	0.15 (0.14 to 0.15)

Note: Null model Brier score = 0.18.

Abbreviation: AUC, area under the receiver operating curve.

TABLE 3 Algorithm performance in independent testing set (95% confidence interval), $n = 222$

Metric	Stochastic gradient boosting	Random forest	Support vector machine	Neural network	Elastic-net penalized logistic regression
AUC	0.72 (0.64–0.80)	0.71 (0.62–0.79)	0.65 (0.56–0.74)	0.77 (0.69–0.83)	0.77 (0.70–0.84)
Calibration intercept	0.01 (−0.34 to 0.35)	−0.02 (−0.37 to 0.33)	−0.01 (−0.34 to 0.32)	−0.04 (−0.39 to 0.31)	−0.02 (−0.37 to 0.32)
Calibration slope	0.83 (0.49–1.17)	0.66 (0.37–0.95)	0.91 (0.45–1.37)	0.90 (0.57–1.22)	1.15 (0.73–1.56)
Brier score	0.16 (0.13–0.19)	0.16 (0.13–0.19)	0.16 (0.13–0.19)	0.15 (0.12–0.19)	0.14 (0.12–0.18)

Note: Null model Brier score = 0.18.

Abbreviation: AUC, area under the receiver operating curve.

VAS Pain, means (IQRs) were 39.8 (34.9–44.9) points, 50.8 (43.5–6.0), and 6 (5–8) points, respectively. Across the entire cohort, 385 (72.8%) patients showed improvement between preoperative and 1-year KOOS following TKA sufficient to meet the MCID. Variables identified to be most predictive of achieving 1-year KOOS MCID were: preoperative KOOS, VAS Pain, preoperative opioid use, PROMIS Global MH, age, and sex. Increasing VAS Pain, PROMIS Global MH, and age predicted increased likelihood of achieving 1-year KOOS MCID whereas increasing preoperative KOOS predicted decreased likelihood of 1-year KOOS MCID achievement. No preoperative opioid use and female sex increased the likelihood of 1-year KOOS MCID achievement.

3.2 | Development and performance of the machine learning algorithms

In cross-validation of the training set, the five models assessed showed comparable discrimination, with an area under the curve (AUC) of 0.74 (95% confidence interval [CI] 0.71–0.77) for stochastic gradient boosting, 0.74 (0.72–0.77) for random forest, 0.75 (0.72–0.77) for support vector machine, 0.77 (0.74–0.79) for neural network, and 0.76 (0.74–0.79) for elastic-net penalized logistic regression (Table 2). Calibration was found to vary between models, with intercepts ranging from −0.28 for the support vector machine to 0.22 for the random forest, and slopes ranging from 0.81 for the random forest to 1.26 for the support vector machine. Brier scores were 0.15 for the stochastic gradient boosting, neural network, and

elastic-net penalized logistic regression models, 0.16 for the random forest, and 0.14 for the neural network. The null-model Brier score was 0.18.

In the internal validation of the models performed within the testing set, the models showed variation in discrimination, with the support vector machine yielding an area under the curve of 0.65, the random forest 0.71, the stochastic gradient boosting 0.72, and both neural network and penalized logistic regression 0.77 (Table 3). Calibration intercepts ranged from −0.04 for the neural network to 0.01 for the stochastic gradient boosting, and slopes ranged from 0.66 for the random forest to 1.15 for the penalized logistic regression. Brier scores were 0.16 for the stochastic gradient boosting, random forest, and support vector machine, 0.15 for the neural network, and 0.14 for the penalized logistic regression. The null-model Brier score was 0.18 for the testing set.

The elastic-net penalized logistic regression model was selected as the best performing model, and the relative importance of factors identified as most important for predicting improvement meeting the MCID on the KOOS at 1-year follow-up was compared within this model (Figure 1). Partial dependence plots showing the relationship between each predictive continuous variable and predicted probabilities for the model were constructed (Figure 2). Calibration plots indicated good calibration for the penalized logistic regression model, while decision curve analysis showed the selected model's predictions to yield greater net benefit than did those of both the default strategies of changing management for all patients or no patients. The selected model also showed a greater net benefit than a model based exclusively on preoperative KOOS (Figure 3).

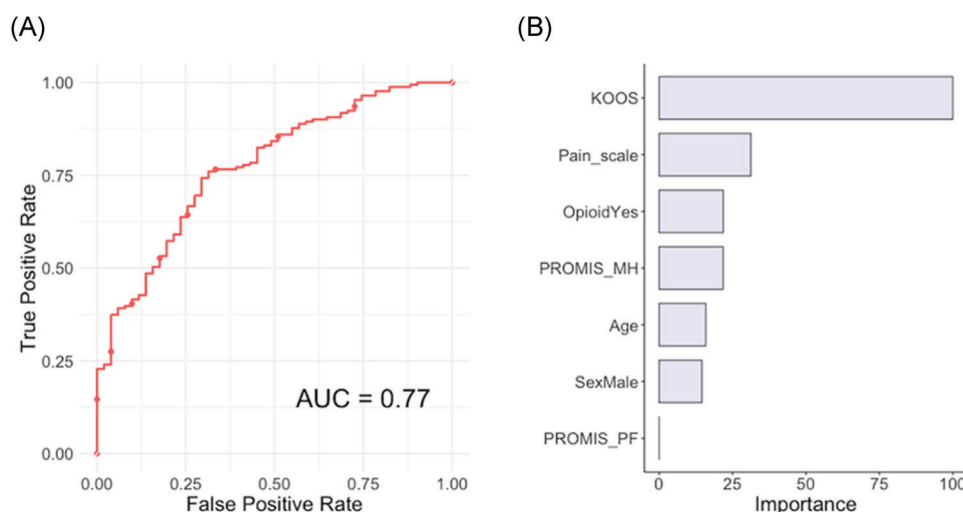


FIGURE 1 (A) C-statistic (area under receiver operating curve) for elastic-net penalized logistic regression and (B) relative variable importance for prediction of KOOS-PS MCID achievement, testing set $n = 222$. KOOS-PS, Knee Injury, and Osteoarthritis Outcome Score-Physical Function Short Form; MCID, minimal clinically important difference [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jor.25125)]

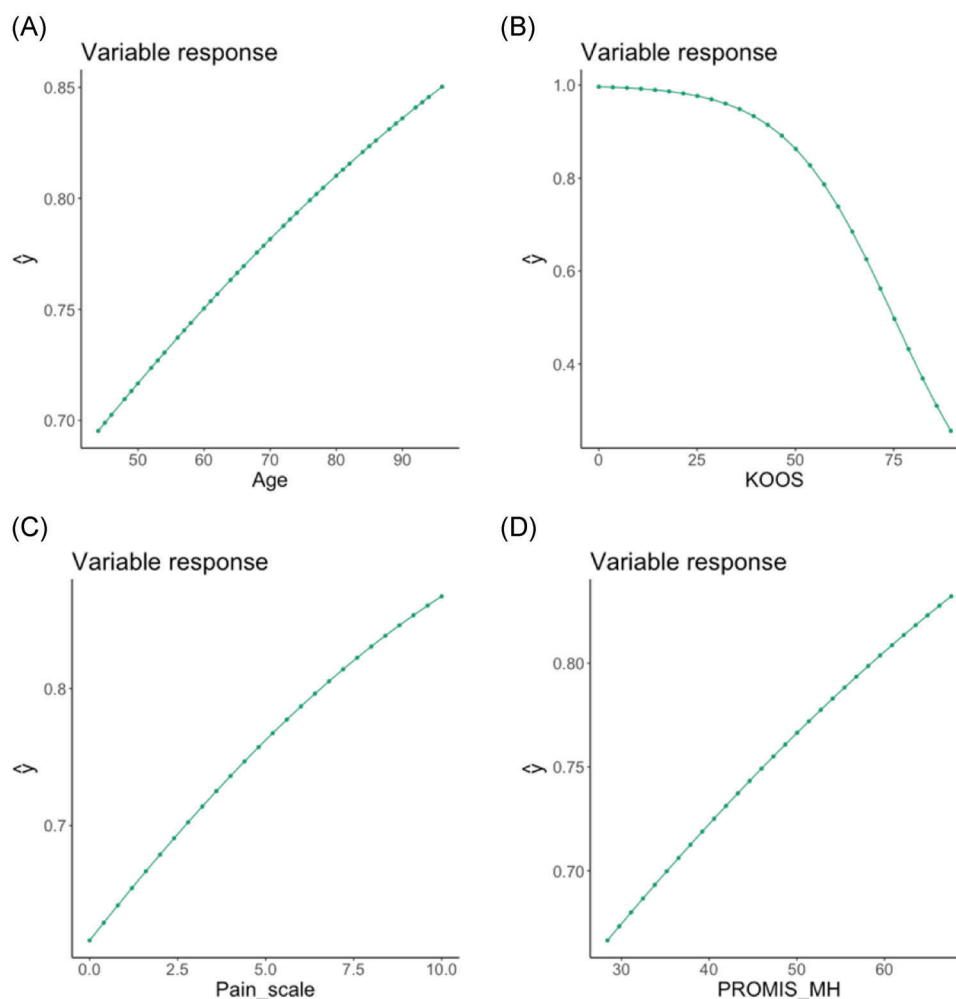


FIGURE 2 Partial dependence plots for continuous variables. (A) Age, (B) baseline KOOS-PS score, (C) VAS Pain scale, (D) baseline PROMIS Mental Health score) predictive of KOOS-PS MCID achievement. KOOS-PS, Knee Injury, and Osteoarthritis Outcome Score-Physical Function Short Form; MCID, minimal clinically important difference; PROMIS, Patient-Reported Outcomes Measurement Information System; VAS Pain, visual analog scale for pain [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jor.25125)]

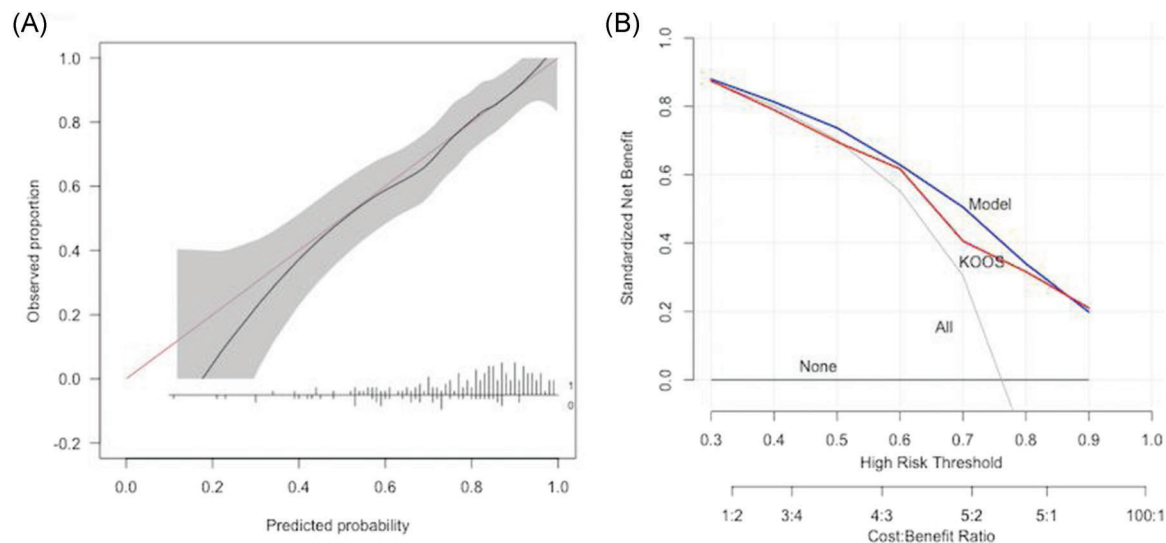


FIGURE 3 (A) Calibration and (B) decision curve analysis of elastic-net penalized logistic regression, testing set $n = 222$ [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

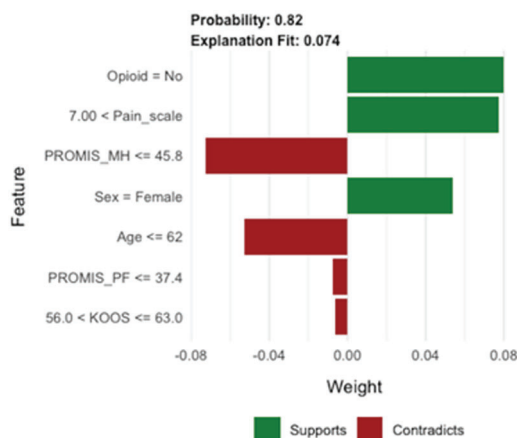


FIGURE 4 Example of individual patient-level explanation integrated into a digital application. PROMIS-PF, Patient-Reported Outcomes Measurement Information System-physical function [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

The selected elastic-net penalized logistic regression model was deployed in an open-access web-based application for individual patient-level predictions, which can be found at https://sorg-apps.shinyapps.io/tka_koos_mcid/. Figure 4 shows an example of predictive results from the model indicating a probability of 82% for an individual patient to meet the MCID between preoperative and 1-year KOOS, as well as the importance of each variable in making this particular prediction.

4 | DISCUSSION

In our analysis, the elastic-net penalized logistic regression model performed well on discrimination, calibration, and decision curve analysis on the testing set for predicting patients likely to achieve

the MCID on the KOOS-PS assessment. Preoperative KOOS-PS score, VAS Pain, PROMIS global mental health score, and preoperative opioid use were found to be the most important risk factors for prediction. 72.8% of patients achieved the MCID on the KOOS-PS following TKA.

This proportion of patients in our study who achieved the KOOS-PS MCID is similar to other studies investigating this metric. Fontana et al.¹ observed an MCID achievement rate of 82% among their cohort of 6,205 patients who completed the KOOS Jr. assessment. This discrepancy can partially be attributed to the longer follow-up period of 2-years chosen by their group. Lyman et al.¹⁷ observed the substantial clinical benefit across various KOOS domains to be between 60% to 78% for patients undergoing TKA. Liddle et al.¹⁸ performed a propensity score-matched study between patients undergoing either a uni-compartmental knee arthroplasty or TKA, and observed that 86.8% of TKA patients achieved the MCID on the Oxford Knee Score.

The factors identified in this study for the prediction of KOOS-PS MCID were consistent with studies that have investigated risk factors associated with adverse PROMs following TKA. Fontana et al.¹ similarly identified baseline KOOS Jr. score as predictive of achieving KOOS Jr. MCID using a machine learning-based approach. Male sex and age under 65 years old have previously been identified as predictive of worse outcomes on pain and function assessments, including the Oxford Knee Score and EuroQol.^{19,20} Smith et al.²¹ similarly reported preoperative opioid use as predictive of a less reduction on the WOMAC pain score following TKA. Our finding that preoperative elevated pain scores were associated with a greater likelihood of achieving the KOOS MCID was in accordance with other studies that have shown greater preoperative satisfaction or function can preclude patients from achieving the MCID; Berliner et al.²² identified a preoperative threshold of 58 on the KOOS with the finding that patients above this threshold were less likely to achieve the MCID compared to patients below this threshold.

Interestingly, decreasing preoperative PROMIS-MH and PROMIS-PF scores were identified as predictive of failure to achieve KOOS-PS MCID in our model. These findings could suggest that for joint-specific PROMs such as the KOOS, a worse preoperative level of functioning permits more potential for improvement, but for full-body, general health questionnaires including PROMIS, achievement of the KOOS-PS MCID is prevented by non-joint related ailments which complicate the perception of an improvement in knee function or pain. Further studies should be conducted to explore the complex relationship between joint-specific and full-body patient-reported outcome measures. The purpose of this study was to provide clinicians with a tool that can aid their individual decision-making to preoperatively assess a particular patient's probability of achieving an acceptable subjective outcome on the KOOS-PS following TKA. The predictive algorithms developed in this study can be further used as a screening tool to determine which patients may perceive their knee function to be improved to a discernable level, and thus are good candidates for surgery.

This study is not without several limitations. First, the analysis performed in this study only includes internal validation via assessment in the testing set. Future studies need to be conducted to see whether the model developed in this study can be applied to different cohorts of patients on external validation. Furthermore, this analysis was retrospective in nature and future prospective studies should be conducted to confirm the relationship between the predictive variables identified by our algorithm and the achievement of KOOS-PS MCID. The approach chosen to obtain the MCID, while validated in previous studies, precludes usage in future studies, thus limiting its generalizability and use as a reference. Preoperative opioid use was identified as a risk factor; unfortunately, our study could not account for variance between providers for opioid prescriptions. Finally, the model only accounted for variables that are routinely collected at researchers' institutions; we acknowledge that there could be other variables collected by other healthcare systems that may be associated with our outcome of interest, however, could not be investigated for the purposes of this study.

5 | CONCLUSION

This study is the first attempt to predict the probability of achieving the MCID of KOOS-PS following TKA using a machine learning-based approach. The results were used to develop a clinical decision aid based on commonly collected predictive variables to preoperatively predict an individual patient's likelihood of achieving the KOOS-PS MCID within 1 year. By preoperatively identifying patients who have a poor chance of achieving a perceptible improvement in knee function following TKA, orthopedic surgeons can counsel these patients accordingly and target modifiable risk factors that may improve their chances of obtaining a subjective positive outcome.

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CONFLICT OF INTERESTS

The authors declare that there are no conflicts of interest.

AUTHOR CONTRIBUTIONS

Aditya V. Karhade, Akhil Katakam, Antonia F. Chen, and David Shin contributed equally to research design, analysis of data, and drafting of the manuscript. Charles Bragdon, Antonia F. Chen, Christopher M. Melnic, Joseph H. Schwab, and Hany S. Bedair contributed equally to critical revisions of the manuscript as well as approval of the final submission. All authors have read and approved the final submitted manuscript.

ORCID

Akhil Katakam  <https://orcid.org/0000-0003-0774-5702>

Antonia F. Chen  <https://orcid.org/0000-0003-2040-8188>

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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